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(a) Outline the characteristic features of the kingdoms Protocista and Fungi.

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- (b) *Xanthoria* is a lichen that can grow in places where there is a high concentration of sulfur dioxide in the air, for example in towns where homes, factories and vehicles burn fuels.

Fig. 9.2 shows a lichen of the genus *Xanthoria*.

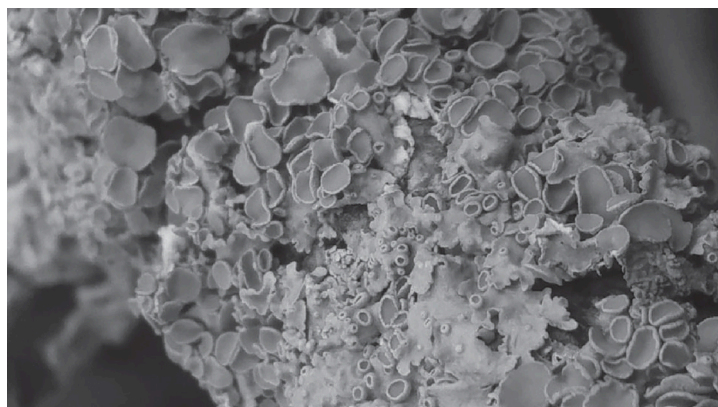


Fig. 9.2

- (i) A student planned a method to measure the relative abundance of *Usnea* and *Xanthoria* on trees along a transect from the town centre at 0 km to unpolluted countryside at 4 km.

Suggest why measuring the relative abundance of the two types of lichen gives information that is useful for conservation.

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- (ii) Although a large biodiversity of lichens can be found in a range of habitats, most people ignore them.

Outline why forms of life that are usually ignored, such as lichens, should be conserved.

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